Krishna Priya R S

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Summary

AI/ML-focused Computer Science undergraduate specializing in Bioinformatics at VIT Vellore. Experienced in developing and deploying deep learning and NLP systems for healthcare and multilingual applications, with strong focus on model performance, evaluation, and real-world deployment.

Education

VIT University, Vellore
Sophomore - B.Tech in Computer Science (Bioinformatics)
Class XII (CBSE) 91.2%
Class X (CBSE) 94%

Technical Skills

Languages: Python, C++, SQL
ML/AI: PyTorch, TensorFlow, Scikit-learn, Transformers, Computer Vision, NLP
Data: Pandas, NumPy, Matplotlib
Tools: Git, Jupyter Notebook, Streamlit

Projects

RetinaScan AI – Multi-Class Retinal Disease Classification
Python, PyTorch, Computer Vision, Streamlit

- Built a deep learning system for retinal disease classification using OCT images.
- Implemented transfer learning using ResNet-18 for multi-class classification (CNV, DME, Drusen, Normal).
- Trained on Kermany 2018 OCT dataset (84K+ images) with preprocessing and augmentation.
- Evaluated model using confusion matrix and loss curves for performance analysis.
- Deployed as an interactive Streamlit web app for real-time image-based diagnosis.
- **Live Demo:** retinascanai.streamlit.app

Multilingual Hate Speech Detection using mBERT
Python, Transformers, NLP

- Fine-tuned multilingual BERT for hate speech detection across Tamil, Bengali, and English datasets.
- Achieved F1-score of 1.00 for hate class and 0.81 for non-hate class (Tamil dataset).
- Reached accuracies of 91% (Tamil), 75% (Bengali), and 97% (English).
- Analyzed classification bias using confusion matrices and improved handling of code-mixed inputs.
- Contributing to an ongoing research paper targeting publication.

Cancer Drug Response Prediction using Gene Expression
Python, Machine Learning, Bioinformatics

- Built a regression model to predict IC50 drug response using gene expression data.
- Simulated high-dimensional genomic datasets (100 features, 200 samples).
- Applied Orthogonal Matching Pursuit (OMP) for sparse regression and feature selection.
- Leveraged PyCaret AutoML for model selection and optimization.
- Achieved R^2 score of 0.97 and deployed via Streamlit for real-time predictions.

Experience

Machine Learning Intern – Shadow Fox
Dec 2025 – Jan 2026

Offer Letter: View Document

- Developed machine learning models on structured datasets with preprocessing and feature engineering.
- Evaluated model performance and implemented validation techniques.
- Delivered end-to-end ML workflows under project constraints.